Agents

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Introduction

- Before understanding knowledge representation, we need to know several things, e.g.
  - Intelligent Agents
  - Agent Behaviour
  - Environments
  - Nature of Environment
  - Structure of agent

Agent

- Agent: is something that acts
- Performance measure: subjective measure to characterize how successful an agent is. It can be measured in terms of:
    - Speed
    - accuracy or quality of the solutions achieved by the agent
    - Power consumption
    - cost, etc...

Agent Types

- Types of agent:
  - Human
    1. eyes, ears, skin, taste bud, etc. that acts as sensors
    2. hand, fingers, legs, mouth, etc. for effectors.
  - Robot
    3. camera, sonar infrared, etc. that acts as sensors.
    4. grippers, wheels, lights, speakers, etc. for actuators.
  - Autonomous Spacecrafts, etc..
  - Expert systems like Cardiologists.

Agent Properties

- Properties
  - Sensing
  - Acting
  - understanding, reasoning, learning

what agent percepts is converted into actions is defined by a function called as agent function:

$f: P^* \longrightarrow A$

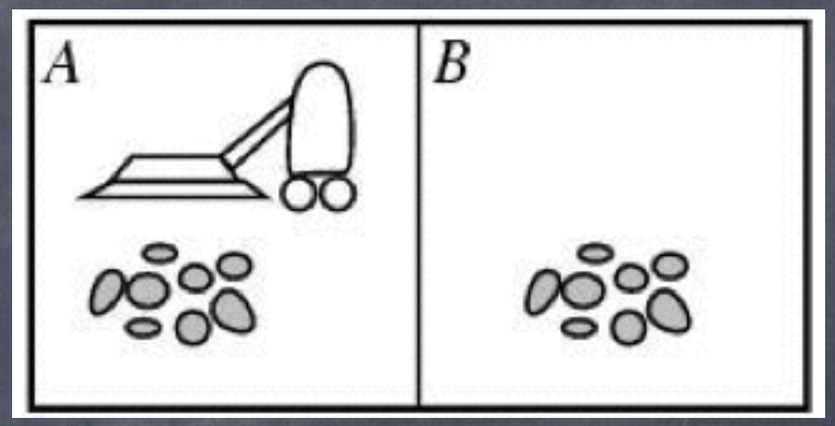
Intelligent Agent

- Intelligent Agent: must sense, must act, must be autonomous (to some extent),
- Must be rational.
- A rational agent always does right thing.
- rational agent is not omniscient. It does not know the actual outcome of its action, and it may not know certain aspects of its environment.
- So Rationality must take into account the limitation of agent.

Example of Agent

- Automated Taxi driving system
  - Percepts: video, sonar, speedometer, GPS, etc..
  - Actions: Steer, Accelerate, brake, horn, etc...
  - Goals: maintain safety, Reach destination, maximise profit (fuel, tire wear), obey laws, provide passenger comfort, etc..
  - Environment: Urban Streets, traffic, pedestrians, weathers, etc..

Example of Agent



- VACCUM CLEANER WORLD
  - Environment: square A and B
  - Percepts: [location and content] e.g. [A, Dirty]

Percept sequence	Action
[A,Clean]	Right
[A, Dirty]	Suck
[B, Clean]	Left
[B, Dirty]	Suck
.....	.....

  - Actions: left, right, suck, no operation

Example of Agent

- Question
- Find About Mars Rover: (A Mars rover is an automated motor vehicle which propels itself across the surface of the planet Mars upon arrival.)
  - What are the percepts for this agent?
  - Characterize the operating environment
  - What are the actions the agent can take?
  - How can one evaluate the performance of the agent?

Environment

- To design a rational agent we must specify its task environment.
- Task environment means: PEAS description of Environment:
    - Performance
    - Environment
    - Actuators
    - Sensors

Types of Environment

- Fully Observable / Partially Observable
  - If an agent's sensors give it access to the complete state of the environment at each point in time.
- Deterministic / Stochastic
  - If the next state of the environment is completely determined by the current state and the action executed by the agent
- Episodic / Sequential
  - In an episodic environment, the agent's experience is divided into atomic episodes. Each episode consists of the agent perceiving and then performing a single action.

Types of Environment

- Static / dynamic
  - If the environment can change while an agent is deliberating, then we say that the environment is dynamic for that agent.
- Discrete / continuous
  - Chess game has finite number of distinct states. Taxi driving is a continuous state and continuous time problem
- Single Agent / Multiagent
  - An agent solving a crossword puzzle by itself is single agent environment. Agent playing chess is in a two agent environment.

THANKS